MIS 201 · WEEK 7 · WEDNESDAY 11/4

Lab 7: The Bellingham Pitch

Recap, four toys, Waymo’s data, the pitch math, then the debate. Pairs, one machine.

Today

2:00	**Recap of Monday**	4 min
2:04	**Lab 7, sections 1 to 4: the four toys**	36 min
2:40	**Section 5: Waymo's data, then the pitch math**	20 min
3:00	**The records: drivers, safety, jobs**	8 min
3:08	**Debate: prep and rounds (follow the Gilman Rules!)**	37 min
3:45	**Wrap-up and attendance**	5 min

DUE FRIDAY 11/6, 11:59 P.M.

- Lab 7 writeup, five items
- Quiz on Chapter 7

Sections 6 and 7 and the writeup are for after class. Jot writeup notes as you go today.

Recap from Monday

1. **The loop**
   Sense, perceive, predict, plan, act. Naming and guessing are the hard steps.
2. **Neural networks**
   A stack of dials tuned on examples. Can’t say why; only knows what it’s seen.
3. **The score sheet**
   It learns what you scored, not what you meant.
4. **The long tail**
   Rare stuff is where it breaks.
5. **Where the human stands**
   In, on, or out of the loop. The middle seat is a trap.
6. **Who’s counting**
   Waymo’s safety numbers are Waymo’s numbers.

WHAT DO YOU THINK?

Tell the person next to you, in one sentence, why “right 99.99 percent of the time” isn’t a finish line. Go.

Today: four toys, Waymo’s own data, the pitch math with Claude, then debate Seattle.

Today you work for the robotaxi company

YOUR ROLE

YOUR BOSS WANTS

To launch in Bellingham.

It's 2028, and made up: each city can permit robotaxis. You need 4 of the council's 7 votes.

THE COUNCIL WILL WEIGH

Riders against drivers.

Crashes against paychecks. Ignore either side and you lose the vote.

YOUR JOB TODAY

Open the car up.

Train it, watch it see, set its score sheet, check Waymo's data, do the math.

Every toy ends with a "For the pitch" box. Each one is something you could offer the council, and what it costs you.

Section 6, at home: the council hearing. Win four votes and keep year one above $1 million.

Four questions running through every section

1. **Can anyone explain a mistake?**
   Section 1. You can see every weight, and you still can't say why it was wrong.

2. **Does it know where it's allowed to be confident?**
   Sections 2 and 4. Rain, fog, snow, and the edge of what it has practiced.

3. **Who keeps the score?**
   Section 5. Waymo's own data, counted and published by Waymo.

4. **Who answers when it's wrong?**
   Section 3. The score sheet you write.

RESPONSIBLE AI · SECTION 2.8

Benefits that are real, risks that are known and limited, and someone who answers when it fails.

Look for the yellow-tag cards. In section 6, the council asks all four.

Part 1: the lab

56 MIN · PAIRS

1	**Mini-Perception: teach it to see**	12 min	done by 2:16
2	**Mini-Camera: the convolutional filter**	6 min	done by 2:22
3	**Mini-Planner: teach the car to turn**	12 min	done by 2:34
4	**Mini-World: the world model**	6 min	done by 2:40
5	**Check Waymo's data, then the pitch math**	20 min	done by 3:00
6	**The council hearing**	after class	*at home*
7	**Under the hood**	after class	*at home*

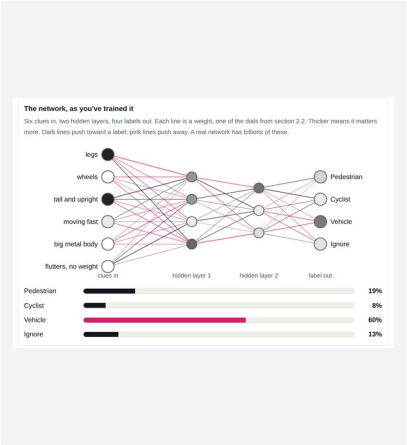
HOW IT WORKS

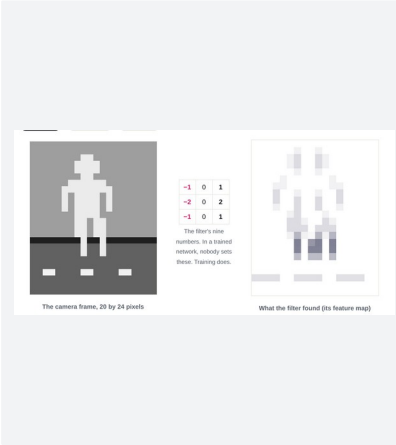
- Pairs, one machine.
- Sections in order; each tool tells you the next step.
- Jot writeup notes as you go.
- Rather not use AI? Section 5 has a spreadsheet path.
- Don't paste anything private into Claude.
- Log out of Claude when you're done.

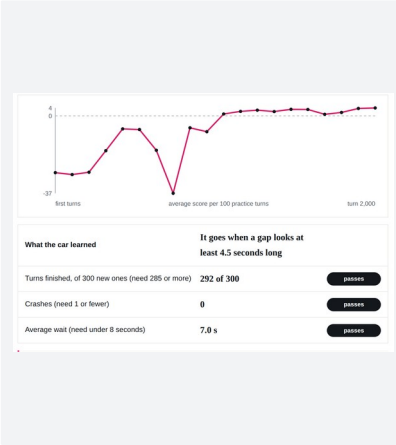
1 and 2 are sections 2.2 and 2.3. 3 is 2.4's reward function. 4 is 2.5's world model. 5 is the case's Questions 2 and 4.

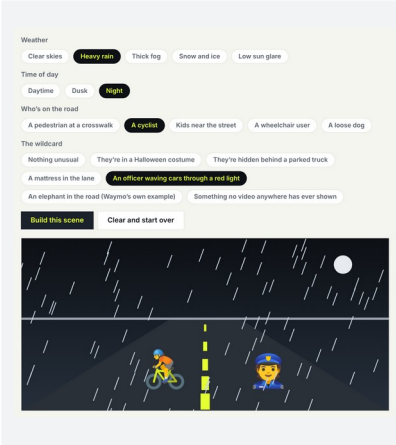
The four toys, and what to watch for

SECTIONS 1 TO 4









1. TEACH IT TO SEE

It's confidently wrong on things it never saw.

2. HOW IT READS A FRAME

Add fog. Watch the verdict fall apart.

3. TEACH IT TO TURN

The chapter's sheet passes. The other two presets each fail a different check.

4. LET IT PRACTICE

Build three scenes, one as weird as you can. Then try the last wildcard.

Section 5, Part A: meet Waymo's data

1. **Phoenix avoids about 14 crashes, San Francisco about 60. Why?**
   Click both cities. Watch which bar moves.

2. **Which crash type improved the least?**
   Scroll to the crash-type chart. Why might that one lag?

3. **Which city's numbers should you trust least?**
   Check how many miles sit behind each city.

4. **What does Bellingham have that none of these cities do?**
   Think weather. Then think about what the data can't see.

5. **Who counted, and who checks?**
   Both bars: Waymo's crashes and the human benchmark.

HOW TO READ IT · 7 MIN

Crashes per million miles. Gray bar: human drivers. Pink bar: Waymo. The gap, times the miles you drive, is crashes avoided.

Answer with your partner before you open the answer. Writeup item 3 asks for two.

Part B: check Claude's math

13 MIN · CLAUDE

AT THE STARTING VALUES: 10 MILLION MILES · 100% TRUST · 15 MILES PER DRIVER HOUR · $20 AN HOUR

Injury crashes avoided per year	3.77 − 0.67 = 3.10, × 10	about 31
Serious-or-worse crashes avoided	0.21 − 0.01 = 0.20, × 10	about 2
Driver hours replaced	10,000,000 ÷ 15	about 666,700
Full-time driver jobs	666,700 ÷ 2,000	about 333
Driver earnings replaced	666,700 × $20	about $13.3M

NOW MOVE TWO THINGS

Which column moved?

Phoenix drops crashes avoided to about 14; San Francisco, about 60. Trust at 50% halves the crash column. The driver column never moves.

Gray rows start from Waymo's data. Pink rows are the analyst's guesses. The page looks equally confident about both.

Download your page before you close Claude. It's writeup item 3, and you'll want it open for the debate and the hearing.

SECTION 8 · THE DEBATE

Badge off

Next: this week's case, the three records, the Gilman Street rules, then your side. Left half of the room is Side A, right half is Side B.

Leave your company badge at the door. In the debate, you speak for Seattle.

This week's case: Waymo in Seattle

CASE OF THE WEEK

Fall 2025

Waymo starts testing in Seattle, with a human behind the wheel. Washington law doesn't allow paid driverless rides.

2026 session: one bill would require a human in every robotaxi; another would allow driverless rides. Neither passed.

October 2025: drivers and Teamsters rally against Waymo.

SIDE A USES IT

Waymo reports far fewer crashes than human drivers, and Seattle still has a human in every driver's seat.

SIDE B USES IT

Those drivers organized, won strong pay protections, and showed up to say no.

Sources: KING 5; Washington State Legislature, HB 1984 and SB 6243; KIRO 7. Full citations on the lab page.

The human driver record

RECORD · 1 OF 3

WHEN		WHAT THE RECORD SHOWS	THE CATCH
2019–20	**Uber's own safety report**	3,824 reports in its five most serious categories of sexual assault on U.S. rides. Riders were the accused party in 43 percent.	*Uber's own count, several years old, and assault is underreported everywhere.*
2023	**Drunk driving**	12,429 people died in crashes with an alcohol-impaired driver: 30 percent of U.S. traffic deaths.	*Most were in their own cars, not taxis.*
2025	**The country**	An estimated 36,640 people died in U.S. motor vehicle crashes, about a hundred a day. Down from 2024.	*An early estimate (NHTSA).*
2025	**Seattle**	30 killed and 221 seriously injured in traffic, most of the dead pedestrians struck by drivers. The goal: zero by 2030.	*All drivers, not rideshare. Nobody has counted taxi or Uber crashes.*

Notice the pattern: huge and steady. Side A leans on this. Side B says most of it happens in private cars a robotaxi never touches.

Robotaxis on the road: the safety numbers

RECORD · 2 OF 3

WHEN		WHAT THE RECORD SHOWS	THE CATCH
Oct 2023	**Cruise loses its permit**	A Cruise robotaxi dragged a pedestrian about 20 feet after a hit-and-run. California suspended its driverless permits.	*GM's Cruise, not Waymo. And the regulator acted.*
Dec 2024	**The insurer**	Swiss Re: 92 percent fewer bodily injury claims over 25.3 million Waymo miles.	*Released jointly with Waymo.*
Dec 2025	**The trauma surgeon**	A trauma surgeon who reviewed the data called self-driving cars a public health breakthrough.	*One doctor's reading of the company's data.*
Jan 2026	**The child near a school**	A Waymo struck a child near a Santa Monica school at drop-off. Minor injuries. Federal regulators opened an investigation.	*Both sides use it. The 14 mph is Waymo's model; regulators ask why it was going 17.*
Spring 2026	**Who feels safe**	USC survey, L.A. County: women feel a lot safer from harassment in a Waymo than in an Uber or Lyft.	*Only 9 percent rated robotaxis safe from crashes.*
To June 2026	**Waymo's own count**	271.3 million driverless miles: 82 percent fewer injury crashes, 95 percent fewer serious-or-worse.	*Waymo's analysis, in five cities, none as wet as Seattle.*

Notice the pattern: every big crash number comes from Waymo or was released with Waymo. Side A leans on the numbers. Side B asks who counted.

Who drives for a living: the jobs numbers

RECORD · 3 OF 3

WHEN		WHAT THE RECORD SHOWS	THE CATCH
2021	**Who drives**	Pew: 5 percent of U.S. adults have ever driven for a ride-hailing app. 42 percent of lower-income gig workers say it's their main job.	*From 2021, and many drive part time.*
2022	**Washington's driver law**	HB 2076: minimum pay per trip, paid sick leave, workers' comp. Supporters called it the highest statewide floor in the country.	*A compromise: drivers stayed contractors.*
2024–34	**The federal forecast**	BLS: about 447,900 taxi, ride-hailing, and chauffeur jobs, projected to grow 9 percent.	*A projection, lumped with shuttle and limo work.*
Late 2025	**Paychecks in robotaxi cities**	Gridwise: trips per hour fell 5.3 percent in robotaxi cities vs. 2.6 nationally; hourly pay fell in L.A., S.F., and Phoenix.	*S.F. drivers earned 16 percent more per quarter. Gridwise can't isolate robotaxis.*
2025–26	**Seattle's drivers say no**	Drivers and Teamsters rally against Waymo. The Washington Drivers Union says about 30,000 people drive rideshare in the state.	*The union's estimate, counting part-timers.*
July 2026	**A middle path**	A D.C. bill would cap robotaxis at 200 cars until 2028 and charge 15 cents a mile.	*Not passed, and unions still oppose it.*

Notice the pattern: job loss shows up first as fewer trips an hour, not pink slips. Side B leans on this. Side A points at the forecast and the middle path.

Debate rules: the Gilman Street rules

APPLY THESE TODAY

Same rules as always, borrowed from 924 Gilman Street. Here's what each one means in today's debate.

NO RACISM · NO SEXISM · NO HOMOPHOBIA · NO TRANSPHOBIA	Argue with ideas, never at people. Extra care today: some of you have family who drive for a living, or have lost someone in a crash.
NO FIGHTING	No attacking a position without evidence. "That's wrong" isn't a point. "That's wrong, because NHTSA's 2025 estimate…" is.
NO STAGEDIVING	No interrupting, piling on, or ambushing. One speaker at a time; let them finish.
NO MAJOR LABEL BANDS	No blind love or hate for a company or a profession. Waymo, Uber, unions, drivers: give reasons, not fandom.

The debate: robotaxis in Seattle?

37 MIN

The motion: **Seattle should let driverless robotaxis take over its taxi and rideshare work.**

SIDE A · LEFT HALF OF THE ROOM

Yes: the safety numbers win

Seattle lets robotaxis carry paying riders with no driver, and lets them grow as fast as riders choose them.

Prove: that the safety numbers are worth the paychecks they cost, though the company counted most of them, in drier cities.

SIDE B · RIGHT HALF OF THE ROOM

No: the drivers' jobs win

Seattle keeps a person in every driver's seat. Robotaxis can keep testing, with a safety driver aboard.

Prove: that the jobs are worth the crashes human drivers will keep causing, and what Seattle does about those crashes instead.

Resolve it: name the one number that would make you say yes, the one that would make you say no, and who should keep the score.

Side A: the safety numbers win

SIDE A

1	**“Human drivers kill people every single day.”**	36,640 U.S. road deaths in 2025. Seattle: 30 killed, 221 seriously injured. 30 percent of 2023 deaths involved a drunk driver.
2	**“The numbers say robotaxis crash far less.”**	Waymo: 82% fewer injury crashes, 95% fewer serious-or-worse, over 271.3M miles. Swiss Re: 92% fewer injury claims.
3	**“The doctors who treat crash victims are on our side.”**	A trauma surgeon who reviewed the data called it a public health breakthrough.
4	**“No driver means no stranger in the car.”**	Uber: 3,824 serious sexual assault reports in 2019–20. USC 2026: women felt much safer from harassment in a Waymo.
5	**“Driving jobs are growing anyway.”**	BLS projects taxi, ride-hailing, and chauffeur jobs up 9 percent, 2024–34, about 58,800 openings a year.

Count off 1 to 5 on your side. That’s your point. The comeback to expect is on the lab page.

Side B: the drivers' jobs win

SIDE B

1	**"Thousands of Washington families drive for a living."**	About 30,000 rideshare drivers in the state, per the drivers' union. 42 percent of lower-income gig workers say it's their main job.
2	**"Washington drivers fought for these jobs, and won."**	The 2022 law: minimum pay per trip, sick leave, workers' comp. The strongest statewide floor in the country.
3	**"It's already hitting paychecks."**	Robotaxi cities: trips per hour down 5.3 percent, twice the national drop. Hourly pay fell in L.A., S.F., and Phoenix.
4	**"The safety record is kept by the company that profits from it."**	Waymo's numbers are Waymo's analysis. When Cruise dragged a pedestrian, the DMV said it learned about it from another agency.
5	**"People don't trust it yet, and a driver does more than drive."**	USC 2026: only 9 percent rated robotaxis safe from crashes. A driver told NBC he'd never have let a rider be harassed the way one Waymo rider was.

Count off 1 to 5 on your side. That's your point. The comeback to expect is on the lab page.

How the debate runs: prep

12 MIN

1 Get your point 1 min

Groups of 2 or 3. Count off 1 to 5 on your side.

2 Check the evidence 3 min

Click your point's tag on the lab page, reread its record, open at least one source.

3 Pressure-test it with Claude 5 min

Copy the prompt on the page, fill in the brackets, send. Check any study it cites against the records. Close it, log out.

4 Write your notes 3 min

In your own words, on paper or in a doc.

YOUR NOTES

1. Your point, in one sentence
2. The evidence and its source
3. The comeback you expect
4. Your answer to it

THE RULE ON AI

Use it to research. Don't read from it.

No AI? Skip step 3. Read two entries in the other side's record and write their strongest line.

How the debate runs: the rounds

25 MIN

Team up	3 min	Join the other groups on your side. Pick your best two points, an opener, and a rebutter.
Openings	6 min	Side A, then Side B. About 45 seconds each: your claim, your evidence, and its source.
Huddle	2 min	Rebutters pick one point the other side made.
Rebuttals	6 min	Side B first. Answer one specific point, with evidence. No new points.
Switch	4 min	What was the other side's best point? Someone new says it in one sentence.
What would change your mind?	4 min	Out loud. Name the specific evidence you'd have to see.

THE GILMAN RULES, APPLIED

- One speaker at a time. Let them finish.
- Every point needs evidence and a source.
- Answer the argument, not the person.
- Reasons, not fandom.

Whoever names real evidence that would move them wins, whichever side they were handed.

BEFORE YOU LEAVE

Upload your notes

Attendance = your notes from today, uploaded to Canvas right after class.

At home: section 6 (the council hearing, about 15 minutes), section 7 (a short read), and the writeup. Item 1 is how it sees; item 2 is how it decides and practices; item 3 is Waymo's data and your pitch math page; item 4 is the debate, with at least one source; item 5 is the hearing: your final offer and vote count.

Friday 11/6, 11:59 p.m.: Lab 7 writeup and the Chapter 7 quiz, both on Canvas.

Log out of Claude before you go. Shared machines.